

ΔKL as an innovation measure for trademark filings: the expected reward to forward flux, and the unexpected reward to backward flux

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Abstract

We compute a Kullback–Leibler surprise difference $\Delta KL = KL_{\text{pros}} - KL_{\text{retr}}$ on each of 10,060,073 U.S. trademark filings in 1990–2021 against per-day exponentially decay-weighted references of the same NICE class (half-life $H = 2$ years; chosen by signal-maximising sweep). Trademark filings are a new substrate for the prospective/retrospective surprise framework of Murdock, Allen, and DeDeo (2017): they are fee-bearing, legally consequential, and filed under economic incentive, so the goods/services prose reflects the firm’s deliberate language choice rather than incidental editorial drift. ΔKL is essentially uncorrelated with the dominant alternative innovation proxy — firm-year patent counts, $r = +0.010$ on 174,569 firm-years — and the magnitude of its correlation with 4-year stock excess returns is comparable to (slightly larger than) that of patents: incremental R^2 over controls of +0.0046 for ΔKL vs. +0.0024 for $\log(1 + n_{\text{patents}})$. Including both predictors yields R^2 gain of +0.0069, essentially the sum of the two separate contributions: the two innovation channels are independent and of comparable magnitude.

Where ΔKL aligns with the distinctiveness literature. On debut filings, the response curve of 5-year post-registration survival on the innovation side ($\Delta KL > 0$) is a clean saturating exponential: $P(\text{survived}) = 0.995 - 0.55 e^{-0.063 \Delta KL}$, $R^2 = 0.499$ across 20 binned points. Survival rises with prospective novelty, concave-down, with diminishing returns. This is the empirical content of optimal distinctiveness (Deephhouse, 1999; Zhao, Fisher, Lounsbury, & Miller, 2017) translated from *level* of category-typicality to *directional* vocabulary flux.

Where it diverges. The strict inverted- U prediction of the optimal-distinctiveness literature (U -shaped pay-off on distinctiveness level, with a turning point at high distinctiveness) does not appear: the response curve keeps rising at the positive extreme, with no inflection. More surprisingly, the harvest side ($\Delta KL < 0$) has its own saturating exponential, fitted much more tightly ($R^2 = 0.978$): filings whose vocabulary lies further into the receding/legacy side of the category are not penalised but rewarded, and the reward grows the deeper the harvest. We read this as *increasing returns to late-stage imitation*, not predicted by the distinctiveness literature. Together the two saturating exponentials produce a survival U on ΔKL in the pooled corpus; the $+U$ holds in 32 of 44 NICE classes, only one industry has a significant inverted- U , and 11 are indeterminate. We rule out nine alternative explanations and identify a cluster-viability mediator that absorbs 84% of the quadratic on survival, leaving a residual linear ΔKL effect intact.

An unresolved asymmetry: harvest dominates the public-listing signal. The probability that a filing’s owner ever appears in SEC EDGAR (proxy for IPO/public listing) is, at the filing level, monotone *decreasing* in ΔKL rather than U -shaped: 5.78% at the most-negative ΔKL decile, 4.0% at the flux-neutral middle, 3.49% at the most-positive decile. The harvest tail captures the EDGAR-listing pattern far more cleanly than the innovation tail does. We do

not have a single-mechanism explanation for why late-stage-imitation filings are so much more strongly associated with the public-listing outcome than novelty filings are at the unit of one filing.

A long companion paper ([main.pdf](#)) reports the per-industry detail, the financial-outcomes panel on $\sim 1,500$ SEC-listed firms, the patent-counts crosswalk, the cluster-viability mediation, and the per-NICE-class deep dives.

1 ΔKL : construction, novelty of the substrate, and relation to existing innovation measures

Construction. For each filing i at filing-date t_i in NICE class c , the goods/services term distribution P_i is compared via support-restricted KL divergence to two reference distributions Q_i^{pros} and Q_i^{retr} built by per-day exponential decay-weighting (half-life $H = 2$ years; the signal-maximising choice across a sweep of $H \in \{0.5, 1, 2, 3, 5\}$ years on Class 9) of all other filings in c before and after t_i respectively. $\Delta KL_i = KL_i^{\text{pros}} - KL_i^{\text{retr}}$ is positive when filing i 's vocabulary is unusual against the recent past *and* closer to the near future — the directional resonance signal of Murdock, Allen, and DeDeo (2017) applied to commercially incentivised text. The two component levels are correlated at $\rho = 0.99$ per filing; the difference is the directional residual.

Why trademarks. Trademark goods/services prose is a new substrate for the prospective/retrospective surprise framework, and its properties make it well-suited to the measurement question. Filings are *fee-bearing* (each U.S. class costs the applicant \$250-\$350; abandonment penalties accrue), *legally consequential* (granted marks become enforceable property rights; non-use within five years is grounds for cancellation), and *filed under economic incentive* (firms file when they intend to commercialise, not when they're being literary). The prose therefore reflects deliberate language choice by a commercial actor with real money at stake on the front end and real legal exposure on the back end. Prior applications of the surprise framework (Murdock, Allen, & DeDeo, 2017, on long-form essays; Barron, Huang, Spang, & DeDeo, 2018, on parliamentary speech) have used corpora where the authors faced reputational but not direct economic cost for their linguistic choices. The trademark corpus is the first such application to incentivised commercial text at this scale (10+ million filings).

Relation to patents. The dominant innovation measure in the firm-level empirical literature is the patent count (Hall, Jaffe, & Trajtenberg, 2005; Kogan, Papanikolaou, Seru, & Stoffman, 2017). ΔKL is essentially uncorrelated with firm-year patent counts ($r = +0.010$ on 174,569 firm-years), and the magnitude of its correlation with 4-year excess stock returns is comparable to (and slightly larger than) that of patents. In the firm-year R^2 decomposition for predicting 4-year excess returns over the S&P 500: controls-only $R^2 = 0.0211$; adding $\log(1 + n_{\text{patents}})$ gives $R^2 = 0.0235$ (incremental +0.0024); adding ΔKL alone gives $R^2 = 0.0257$ (incremental +0.0046, roughly twice patents); including both gives $R^2 = 0.0280$ (incremental +0.0069 over controls, essentially the sum of the two contributions). The two innovation channels are independent and of comparable magnitude. Patents proxy a firm's continuing R&D output; ΔKL on the firm's debut filing proxies whether the firm started in fertile linguistic ground.

Relation to the distinctiveness literature, in advance. The closest-adjacent management-theory framework is optimal distinctiveness (Deephhouse, 1999; Zhao, Fisher, Lounsbury, & Miller, 2017), which predicts an inverted- U payoff on the *level* of category-distinctiveness: too typical

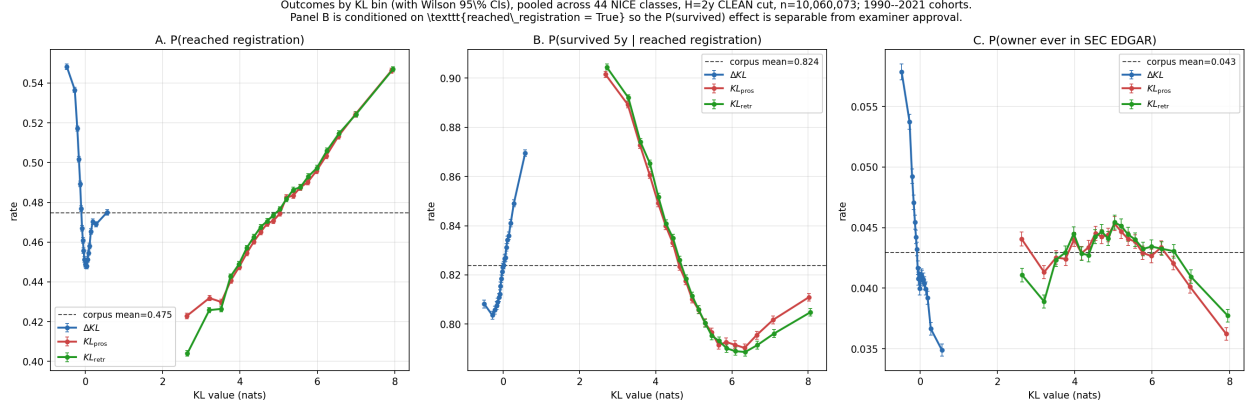


Figure 1: Three outcomes by KL bin (Wilson 95% CIs as error bars), pooled across all 44 NICE classes; $n = 10,060,073$ clean H=2y filings, 1990–2021 cohorts. Each panel shows a different shape on ΔKL (blue). **Panel A ($P(\text{reached registration})$):** *U-shaped* on ΔKL , asymmetric, with the harvest extreme (54.8%) higher than the innovation extreme (47.5%) and the flux-neutral middle lowest ($\sim 45\%$). **Panel B ($P(\text{survived } 5y \mid \text{reached registration})$):** *monotone increasing* in ΔKL — once past the examiner, more prospective novelty means more survival. The substantive post-registration mortality signal favours the innovation side. **Panel C ($P(\text{owner ever in SEC EDGAR})$):** *monotone decreasing* in ΔKL (5.78% at the most-negative decile to 3.49% at the most-positive decile). The public-listing signal is dominated by the harvest tail; we discuss this unresolved asymmetry below. On the component KLs (red, green), registration is monotone-up (more-distinctive marks collide less with prior art and clear examination more easily under 15 U.S.C. § 1052(d)).

is invisible, too distinctive is illegitimate. ΔKL is not a level measure of distinctiveness; it is a *directional* measure of where a filing sits between recent past and near future of the category’s vocabulary. The results that follow align with the empirical content of optimal distinctiveness on the positive- ΔKL side (saturating returns to novelty) but diverge from its strict inverted- U prediction (no turn-down at the extreme) and add a second saturating-exponential response curve on the harvest side that the literature does not predict. We present both as results.

2 Two results: optimal distinctiveness on the innovation tail; increasing rewards to late-stage imitation on the harvest tail

Figure 1 is the central empirical object. The U on *joint* 5-year survival (reached registration *and* still live; not separately plotted but obtained by multiplying Panel A by Panel B per bin) places the flux-neutral middle of ΔKL lowest in the corpus at 37% vs. 44% at the harvest extreme and 41% at the innovation extreme. Both extremes outperform the middle on this joint outcome; the functional form of each tail tells a different theoretical story, to which we now turn.

Innovation tail: optimal distinctiveness as a saturating exponential. On the debut-filings-only panel ($n = 5,088,700$ owner-first filings), we fit a saturating exponential of survival on $|\Delta KL|$ on each side: $P(\text{survived}) = a + b e^{-c|\Delta KL|}$ on 20 binned points. On the positive side ($\Delta KL > 0$) the fit is $a = 0.995$, $b = -0.55$, $c = 0.063$ with $R^2 = 0.499$ across 20 bins. The curve is monotone increasing in ΔKL , concave-down, saturating: an additional unit of prospective-novelty pays at a decreasing rate. This is the empirical content of optimal distinctiveness (Deephouse 1999, *SMJ*; Zhao, Fisher, Lounsbury & Miller 2017, *SMJ*) translated from *level* of distinctiveness to *directional* vocabulary flux: novelty pays with diminishing returns, no inflection where “too

distinctive” becomes a penalty. The strict inverted- U prediction of the original theory is not what we find, but the underlying empirical claim — that distinctiveness has positive but saturating returns — is supported, and the exponential is the explicit functional form.

Harvest tail: increasing rewards to late-stage imitation. On the negative side ($\Delta KL < 0$) the same functional form fits much more tightly: $a = 0.573$, $b = -0.137$, $c = 3.33$, $R^2 = 0.978$ across 20 bins. Survival rises from ~ 0.44 at $\Delta KL \approx 0$ to ~ 0.57 at the most extreme negative ΔKL . Conditional on negative ΔKL the Pearson correlation between $|\Delta KL|$ and survival is $+0.054$ ($n = 2,586,126$ debuts; Spearman $+0.058$). The harvest tail is doing real work and is unmistakably present. The substantive interpretation: filings whose vocabulary lies further into the receding/legacy side of the category are not penalised but rewarded, and the reward *grows* the deeper one harvests. The further past the category’s modal present-day vocabulary a debut filing sits, the more durable its mark turns out to be. This is the late-stage-imitation analogue of optimal distinctiveness: rather than diminishing returns to novelty, the data show *increasing* returns to picking up legacy vocabulary as the field collectively abandons it. The asymmetric saturation rates ($c = 0.063$ on the innovation side vs. $c = 3.33$ on the harvest side) say the harvest side approaches its asymptote much faster, which is consistent with the small-firm-can-pick-up-the-legacy-niche story: there are not that many firms staying in legacy vocabulary; the ones who do are competing against few peers and the marginal return to going even deeper into legacy is high but quickly capped by how much of the niche exists at all.

The most striking single-token examples of the harvest tail are in retreating industries. Negative- ΔKL tokens that survive at 50–80% in the corpus: *microfiche*, *dial-up*, *pager*, *store-and-forward*, *lager*, *stout*, *shandy*, *annuity*, *microcomputers*. None of these are growing categories; all support trademarks that the filing firm keeps alive long after the vocabulary itself recedes from the corpus.

3 How durable is the U ? Industry $\times$ year breakdown

Table 1 reports the U -shape classification of every (industry, year) cell in the panel. For each of 44 industries with $\geq 5,000$ clean filings, we fit a logistic regression of 5y survival on $z(\Delta KL)$, $z(\Delta KL)^2$, and $\log(n_{\text{terms}})$ separately for each filing-year 1990–2021, and classify the cell by whether the quadratic coefficient is significantly positive (the U), significantly negative (inverted- U), or indeterminate.

Where does the U not hold and why? The single significant inverted- U industry is Telecommunications (NICE 038, $n = 170,826$, $t_{\text{quad}} = -11.8$). It has a positive linear ΔKL coefficient ($+0.015$): forward-flux still helps in telecoms, but the harvest tail is absent — old telecom vocabulary does not survive in the way old typesetting or old-beverage vocabulary does. The 11 indeterminate industries split: some (Scientific & Tech Services 042, Transport 039) have a positive linear coefficient on ΔKL but neither a significant quadratic nor a strong harvest tail; others (Hand Tools, Trim, Jewelry, Vehicles, Ropes & Textiles, Musical Instruments) have a *negative* linear coefficient — mild harvest advantage, no significant innovation tail to complete the U . The interpretation in both cases is that the industry advantages flux in one direction but not the other, so the quadratic does not register as significantly U -shaped. None of the 11 are inverted- U industries where firms in the flux-neutral middle outperform both extremes.

Statistic	count	% of total
Industries (of 44 with sufficient data)		
with significant $+U$ on ΔKL	32	73%
with significant inverted- U on ΔKL	1	2%
indeterminate	11	25%
Industry $\times$ year cells (of 1,267 total)		
significant $+U$ on ΔKL ($z > +1.96$)	338	27%
significant inverted- U ($z < -1.96$)	81	6%
indeterminate quadratic	848	67%
linear advantage $+\Delta KL$ ($z > +1.96$)	197	16%
linear advantage $-\Delta KL$ ($z < -1.96$)	484	38%
neither (linear $ z \leq 1.96$)	586	46%

Table 1: U -shape classification of 44 NICE classes individually and of the 1,267 (industry, year) cells where a logit fit converged. The pooled $+U$ holds in 32 of 44 industries; only 1 has a significant inverted- U (Telecommunications, where the linear ΔKL effect is positive but the harvest tail is absent). The remaining 11 are not inverted- U industries — they are indeterminate (no significant curvature in either direction). Cell-level: 27% of cells have a significant $+U$ on ΔKL ; 6% have a significant inverted- U ; 67% are indeterminate. Linear-direction counting at the cell level shows a striking asymmetry: 38% of cells have a significant linear advantage to *negative* ΔKL (the harvest direction), versus 16% with significant advantage to positive ΔKL .

4 Nine alternative explanations ruled out; cluster-viability mediates

Table 2 reports the falsification battery in condensed form. The headline result is that nine alternative hypotheses for the U are ruled out and a tenth check (competition-density mediation) identifies the mediator that absorbs 84% of the quadratic on survival, leaving the linear ΔKL effect intact.

The mediator clarifies the substantive interpretation. Filings at flux extremes are in vocabulary clusters whose peers can also support real businesses; the flux-neutral middle is in clusters where peers are also failing. The survival U is a niche-viability signal mediated through ΔKL . The linear residual — “forward flux still helps after the mediator is included” — is the part of ΔKL ’s information that competition-density alone cannot reproduce.

5 What we don’t know

(i) Whether the harvest tail reflects intelligent harvesting (firms recognising an industry transition and extracting from the legacy side with their eyes open), passive late-occupation (firms whose business naturally uses the receding vocabulary because they are still doing the legacy thing), or selection on something unobservable (firms surviving long enough to file legacy-vocabulary trademarks have already proved durable). The trademark record does not distinguish these. One additional confound we cannot fully rule out in principle is that the harvest tail’s category vocabulary recedes in the corpus precisely because a single dominant firm has captured the category — “Xerox” becomes the only trademark using **xerographic** because everyone else exits the niche, not because the niche is dying. On the 1990–2021 window this story is implausible at the scale we observe ($n = 2.6\text{M}$ harvest-side debuts across 44 industries, none of them dominated by a single firm to that degree), but we flag it explicitly in the long version. (ii) Why the linear $z(\Delta KL)$ effect on survival survives the cluster-viability control. The most plausible explanation is that conditional

on a viable cluster, firms whose vocabulary moves toward the cluster’s future are ahead of firms whose vocabulary moves against it. (iii) Whether the firm-level financial reward to flux (+3.2 pp gross margin per σ of trailing ΔKL for public firms; +28 pp 4-year excess return per σ of debut ΔKL) is robust to the cluster-viability control. We have not yet computed the firm-level mediation regression.

The companion paper [paper/main.pdf](#) reports the same headline finding with substantially more apparatus: full per-industry detail (44 NICE classes worked separately), the financial-outcomes panel on $\sim 1,500$ SEC-listed firms, multi-horizon stock returns analysis with patent-count controls, the cross-buzzword pioneer-ladder analysis, the canonical-winner-and-failed-firm KL profile, the per-industry deep-dive appendix with 44 scatter plots of public firms by debut year, and full long-form treatment of all ten falsification tests in Appendix C.

Alternative	Outcome
(1) Foreign-extension artefact (Madrid-Protocol marks)	Rejected. US-only and foreign-only sub-samples give nearly identical quadratics (+0.0078 and +0.0100).
(2) First-time vs. recurring filings	Both have the U ; recurring slightly stronger.
(3) Same U on financial outcomes	No. Firm-year gross margin is roughly linear in ΔKL (+3.2 pp/ σ); the U lives in survival.
(4) Large-incumbent survivor bias on the harvest tail	Partial. The bottom ΔKL decile has larger mean firm filing-volumes than the middle.
(5) “Liquid Death” hypothesis (low- ΔKL is novel products in old vocabulary)	Rejected on text-length splits.
(6) The U is firm-level (firms with flux-neutral filings just fail)	Rejected. Within firms that abandoned some marks and kept others alive (the “mixed” cohort, 349,819 firms), the U on filing-level survival is preserved.
(7) Blockchain / NFT wave as imitated-but-failed counterexample	Rejected. The crypto cohort outperforms the corpus on every survival margin.
(8) Vintage pooling artefact	Rejected. Per-year fits 1990–2021 show the pooled quadratic is significantly positive in 23 of 32 years independently; the U does not track the macro cycle.
(9) English-language usage drift (Google Books n-grams)	Partial shared signal, not a confound. Per-token correlation between trademark log-lift and Google Books English log-lift is +0.47 ($n = 53$). About 22% of trademark-flux variance per token is shared with general English usage; 78% is corpus-specific.
(10) Competition-density mediation (the explanation that sticks)	$\log(1 + \text{live_competitors})$ in an asymmetric $[t - 2y, t + 5y]$ window (cosine ≥ 0.7 on g/s TF-IDF; not the H=2y reference window) absorbs 84% of the ΔKL quadratic on survival. Flux extremes are in vocabulary clusters where peers also survive, not in less-competitive clusters. Linear $z(\Delta KL)$ effect remains positive ($z = +2.65$) after the mediator is included; flux is partly but not wholly redundant with cluster viability.

Table 2: Falsification battery for the survival U on ΔKL . Test (10) is a mediator (it explains most of the shape) rather than a confound (it doesn’t make the shape go away). Long-form treatment of every test is in Appendix C of the companion paper.